# Comment on "A genetic history of the Balkans from Roman frontier to Slavic migrations": the omitted NorthMacedonia_IA substrate in the 1-source qpAdm screen

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**Abstract**

Olalde et al. [1] begin their present-day Balkan qpAdm analysis with a 1-source substrate screen reported in Data S1 §5 and Data S2 Table 8. This comment addresses one narrow methodological point in that screen. We restate it on the exact public AADR Human Origins labels available in the dataset used in this analysis: `Croatian`, `Serbian_Serb`, `Romanian`, `Bulgarian`, `Albanian`, `Greek`, and `Cypriot`. Under the documented setup, all six published local substrate columns fail across this retained panel. We then add one plausible omitted Iron Age Balkan candidate, `NorthMacedonia_IA`, which was publicly available before Olalde et al. [1] and lies in the same broad regional and chronological frame as several published substrates.

The result is straightforward. On this strict public-label panel, every published substrate column remains rejected under `allsnps=TRUE`, whereas the added `NorthMacedonia_IA` column passes all seven retained rows, with p-values ranging from 0.246 to 0.971. It shows that the published first-stage local-substrate screen was not exhaustive even at the simplest 1-source level. For a screening step used to constrain later model choice, one relevant omitted candidate that reverses the pass/fail pattern is sufficient to demonstrate that point.

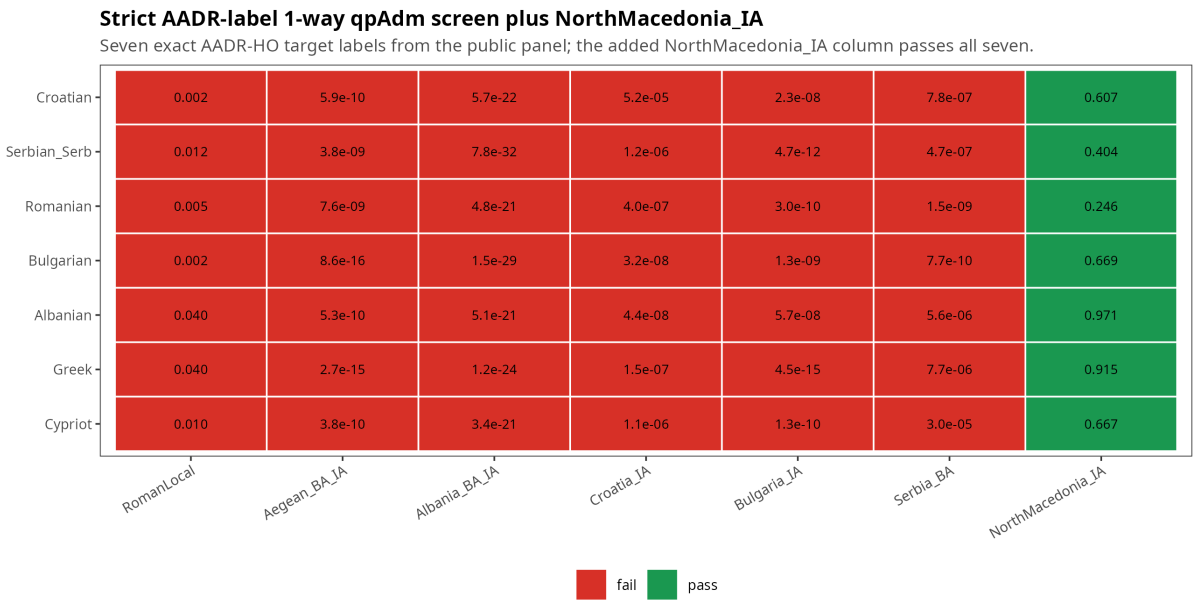


**Figure 1:** Strict AADR-label 1-source qpAdm screen plus `NorthMacedonia_IA`. Cells report qpAdm $p$-values under `allsnps=TRUE`; green indicates non-rejection ($p > 0.05$).

**Table 1:** Strict AADR-label 1-way screen versus the added `NorthMacedonia_IA` column.

| Target | Best published source | Best published $p$ | `NorthMacedonia_IA` $p$ | Verdict |
|---|---|---|---|---|
| Croatian | CroatiaSerbia_RomanLocal | 0.002 | 0.607 | passes |
| Serbian_Serb | CroatiaSerbia_RomanLocal | 0.012 | 0.404 | passes |
| Romanian | CroatiaSerbia_RomanLocal | 0.005 | 0.246 | passes |
| Bulgarian | CroatiaSerbia_RomanLocal | 0.002 | 0.669 | passes |
| Albanian | CroatiaSerbia_RomanLocal | 0.040 | 0.971 | passes |
| Greek | CroatiaSerbia_RomanLocal | 0.040 | 0.915 | passes |
| Cypriot | CroatiaSerbia_RomanLocal | 0.010 | 0.667 | passes |

# 1 Introduction

Olalde et al. [1] model Balkan population history with qpAdm using a fixed right-set and a series of pre-Slavic Balkan substrates. For present-day Balkan and Aegean groups, their workflow begins with a 1-source screen in Data S1 §5 and Data S2 Table 8, followed by target-specific multi-source models. The reproducible step addressed here is that first-stage 1-source screen.

This comment uses a strict public-label test rather than proxy relabeling of every printed Table 8 row. The printed local 1-source rows include `Croatian`, `Serbian`, `Romanian`, `Bulgarian`, `Albanian`, and several regional Greek labels listed in Supplementary Table S2. The analysis therefore stays with the exact public AADR Human Origins labels available in the processed dataset: `Croatian`, `Serbian_Serb`, `Romanian`, `Bulgarian`, `Albanian`, `Greek`, and `Cypriot`. The question is simple: on this strict seven-label public panel, does adding the omitted Iron Age Balkan cluster `NorthMacedonia_IA` materially change the pass/fail pattern?

That omitted cluster matters because it was available well before Olalde et al. [1]. The 14 `NorthMacedonia_IA` individuals were published in Lazaridis et al. [2], the same paper that supplied multiple Iron Age Balkan substrates used by Olalde, and Iosif Lazaridis is a coauthor of both studies. This is therefore not a case of importing a later or peripheral candidate from outside the relevant literature, but of an obvious candidate from the same immediate authorial and evidentiary context being left out of the published stress test.

# 2 Methods

## 2.1 Data and published setup

Genotypes are taken from the Allen Ancient DNA Resource v66.0 Human Origins variant [3]. The present-day analysis considered here is the published 1-source screen documented by Olalde et al. [1] for Balkan and Aegean populations. The documented right-set is: `OldAfrica`, `Steppe_BA`, `EHG`, `Iron_Gates_HG`, `Anatolia_N`, `Iran_N`, `Iberia_IA`, `Greece_Minoan`, `CroatiaMLBA_SloveniaIA`, `Netherlands_MBA_IA`, `Steppe_IA`, `SoutheastTurkey_Byzantine`, and `Baltic_BA`.

The six published source columns are represented here by the public labels `CroatiaSerbia_RomanLocal`, `Aegean_BA_IA`, `Albania_BA_IA`, `Croatia_IA`, `Bulgaria_IA`, and `Serbia_BA`; in Olalde's printed Table 8, the corresponding Bulgaria column is labeled `Bulgaria_EIA`. We reproduce that screen under the same setup and then add `NorthMacedonia_IA` as a seventh, explicitly new column.

The retained target panel is the exact public-HO set used in the pipeline: `Croatian`, `Serbian_Serb`, `Romanian`, `Bulgarian`, `Albanian`, `Greek`, and `Cypriot`. It is therefore a strict public-label panel, not a row-for-row reproduction of every printed Table 8 target. Supplementary Table S2 lists these labels and the exclusion policy.

### 2.2 Implementation

qpAdm models were computed in ADMIXTOOLS 2 [4]. Olalde ran classic ADMIXTOOLS with `allsnps:YES`; in ADMIXTOOLS 2 the closest match is `qpadm(allsnps=TRUE)`, which reads from the genotype files directly rather than from a global $f_2$ cache. The present analysis uses that `allsnps=TRUE` implementation for the published 1-source screen and for the added `NorthMacedonia_IA` column.

A 1-source qpAdm pass is treated here as non-rejection under this right-set and dataset. It is not, by itself, a claim of literal historical ancestry or of robustness to alternative right-sets and source rotations.

## 3 Results

### 3.1 The reproduced 1-source screen still rejects every published substrate

Figure 1 and Table 1 summarize the strict seven-label result. On the retained public-HO panel, every one of Olalde's six published 1-source substrate columns still fails every row under `allsnps=TRUE`. The best published p-value in each row is achieved by `CroatiaSerbia_RomanLocal` and remains below the conventional acceptance threshold, ranging from 0.0016 (`Croatian`) to 0.0401 (`Albanian`).

This reproduces the qualitative point of Olalde's own first-stage screen: none of the published local substrates passes on this strict public-data subset under the documented right-set.

### 3.2 Adding `NorthMacedonia_IA` changes the pass/fail pattern completely

When `NorthMacedonia_IA` is appended as a seventh source column under the same setup, the pattern changes sharply. All seven retained target rows pass, with p-values from 0.246 (`Romanian`) to 0.971 (`Albanian`); see Table 1. This is a material change in the first-stage screen, not a marginal numerical perturbation.

The implication is immediate. Even before any later model choice about Slavic, Anatolian, or Roman-period components, this strict public-label test shows that the published local substrate set was not exhaustive. An Iron Age Balkan substrate absent from the published table is less strongly rejected than all six published candidates and, in this implementation of Olalde's documented setup, is not rejected at all on the retained panel.

## 4 Discussion

The point of this comment is narrow but consequential. In the strict public-label restatement of the published present-day 1-source qpAdm screen, all six of Olalde's published local substrate columns are rejected across the retained target panel, whereas the omitted `NorthMacedonia_IA` column is not rejected in any retained row. That is enough to show that the published screen was not exhaustive with respect to plausible Iron Age Balkan substrate candidates.

Methodologically, that point is straightforward. In Olalde et al. [1], the first-stage screen functions as a filter on later model choice. If a relevant and previously available Iron Age Balkan candidate within the same broad temporal and geographic frame was omitted, and adding it reverses the pass/fail pattern of the screen, then the entry point of the model family depends materially on source choice. The strict public-label panel does not weaken that inference: on the exact public targets analyzed here, the published six-source set fails uniformly while the omitted `NorthMacedonia_IA` source does not. This does not imply that `NorthMacedonia_IA` was a literal ancestor or that a 1-source qpAdm pass establishes continuity. It does show that this source dependence should be recognized explicitly.

## Data and code availability

All analysis code, intermediate results, and the LaTeX source are available at the GitHub repository. Genotype data are not redistributed; researchers must download the AADR v66.0 Human Origins dataset directly from the Harvard Dataverse release.

## References


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